

Midterm Review

The exam will open on Friday, May 8th at 12PM and will be due Tuesday, May 12th at 12PM.

Spring 2026

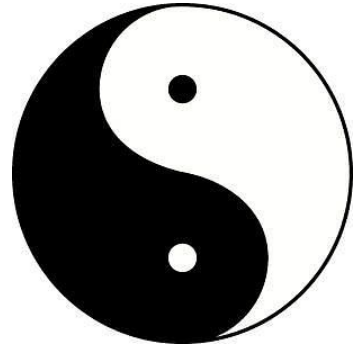
Lectures covered on the midterm

- Lecture 1: Normal Sleep and Dreams
- Lecture 2: Normal Sleep and Dreams II
- Lecture 3: Circadian Sleep Science
- Lecture 4: Sleep and Transportation Safety
- Lecture 5: Dreams and Psychotherapy
- Lecture 6: Dreams and Psychotherapy II
- Lecture 7: Obstructive Sleep Apnea
- Lecture 8: Sleep and Sports
- Lecture 9: Narcolepsy
- Lecture 10: Sleep and Cognition
- Lecture 11: Sleep and Meditation

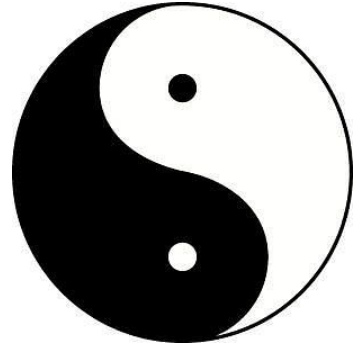
Introduction to Sleep

Sleep Essentials and Class Origins

- Class motto: "Drowsiness is Red Alert"
- Sleep is a rapidly reversible, restorative process
- Sleep reflects your life, involves consciousness
- Sleep supports memory processing and learning
- Sleep is paradoxical



Introduction to Sleep Part II



Sleep Clinic

- Sleep is an active, biologically regulated state
- Sleepiness and sleep timing are driven by biological rhythms
- Sleep has distinct stages with identifiable brain signatures
- EEG measures brain waves, EOG measures eye movements, ECG measures heart rhythm

Circadian Sleep Science

Stages, Cycles, and Sleep Activity

- Circadian rhythms run on an internal clock
- Light is the most powerful tool for shifting circadian timing
- Jet lag caused by circadian misalignment
- Light characteristics: time of day, intensity, duration, color
- Short naps can't fully replace lost nighttime sleep
- Homeostatic sleep pressure decreases while you sleep, increases while awake

Sleep & Transportation Safety (Rosekind)

Guest Lecture

- Fatigue significant cause of accidents, impacts reaction time
- Strategic nap: ideal up to 45 min with 15 min wake-up buffer
- Performance deteriorates before visible signs of sleepiness
- "4 D's" of impaired driving: Drunk, Drugged, Distracted, Drowsy (not Dizzy)
- Performance under sleep deprivation is variable

Dreams and Psychotherapy (Charlton)

Guest Lecture

- Psychodynamic therapy..
 - Emphasizes the unconscious mind, childhood experiences, and the significance of dreams
 - NOT focused on avoiding unconscious conflict
- Dreams are...
 - Consist of visual images often tied to sensory experiences
 - Reflect worries, daily experiences, or fantastical elements
 - Emotionally charged
 - A bridge of the unconscious and conscious awareness
- Interpersonal relationships and attachments are important!

Dreams and Psychotherapy pt. 2 (Charlton)

Guest Lecture

- A dream not interpreted is like an unread letter
- Nightmares are simply a type of dream and not distinct from them
- Kids tend to be more sensitive to nightmares
- REM sleep and dreaming are NOT the same thing

Obstructive Sleep Apnea

- Breathing pauses (>10 s) due to throat blockage
- Symptoms: hypertension, obesity, snoring, daytime sleepiness
- Risks: cardiovascular diseases, mortality increase
- Treatments: CPAP, positional therapy, surgery, oral appliances

Types of Apneas

- Obstructive Sleep Apnea (OSA): airway blockage
- Central Sleep Apnea (CSA): lack of respiratory effort
- Mixed Sleep Apnea: combination of OSA and CSA

Sleep and Sports (Mah)

Guest Lecture

- Sleep is framed as a performance tool, not just recovery
- Athlete sleep is shaped by schedules, travel, substances, and mental load
- Sleep habits matter long before game day
- Movement patterns become more variable with sleep deprivation, increasing injury risk
- Environmental cues (light, heat) and routines influence sleep quality and efficiency
- More sleep can translate into measurable performance gains on the court (*The Effects of Sleep Extension on the Athletic Performance of Collegiate Basketball Players* study)

Narcolepsy (Flygare)

Guest Lecture

- Narcolepsy involves daytime sleepiness and emotion-triggered muscle changes
- Key symptoms - Excessive daytime sleepiness, cataplexy, sleep paralysis
- Sleep habits matter long before game day
- Certain substances and behaviors can worsen symptoms
- Living with narcolepsy involves limited daily energy (spoon theory)
- Rare and socially isolating disease

Sleep and Cognition (Heller)

Guest Lecture

- Circadian misalignment leads to decrease in memory consolidation, emotional stability, and creativity
- Brain States: Wake, REM, NREM
- Memory Types: Declarative, Non-Declarative
- Memory Stages: Acquisition/Encoding, Consolidation, Association/Integration, Recall/Recognition, Reconsolidation, Erasure
- Memory consolidation: “A singular, time-dependent process involving stabilization of learned information”
 - Sleep is the first step
- Essential Neurophysiology:
 - Hippocampus: declarative short term and formation of long-term memory
 - Suprachiasmatic Nucleus: circadian rhythms

Sleep and Meditation (Abrahamson)

Guest Lecture

- Brain development continues into adulthood
- Mental training can change how the brain functions, but doesn't "grow more brain"
- The mind naturally wanders, but modern life makes it hard to rest our brains
- Not all sleep has the same restorative value
- Stress involves coordinated hormonal and physiological responses
- Mindfulness practices engage specific brain networks related to awareness and attention

Tips for Midterm

Study and Ask Questions!

- Go over lecture slides and take notes
- Email any of the TAs if you have any questions beforehand
- Make sure to submit the exam on time. It does not submit itself!
- Scared about your grade? DO THE EXTRA CREDIT OPPORTUNITIES!

And don't forget:

**Drowsiness is
Red Alert**